A NEW CONCEPTUAL RESEARCH NOTE · 8 NOVEMBER 2026

After the Line

Anchored values, signal diffusion, and the recovery of meaning after cognitive collapse

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With drafting, computational, and research assistance from Grok Bot

> **Conceptual, not validated science.** This is an original mathematical design proposal with a toy simulation, not a discovered law of consciousness, a diagnostic instrument, or evidence of treatment effectiveness.

Abstract

What allows a person to carry a remembered value into a next action after a moment of overwhelm? Building on MindFrame, Project Ariel, and Espinosa's signal-transmission model, this paper separates two mechanisms: retaining a chosen anchor and transmitting its influence to other contexts. We propose an anchored reaction–diffusion system with leaky integral memory. A deterministic 21-node simulation compares no anchor, an isolated anchor, and two diffusion rates following a complete signal reset. At time 80, an isolated anchor retains a local signal of 0.938 but produces no signal at a distant output node. Adding diffusion produces distant signals of 0.179 and 0.312 while reducing the local peak. This formal trade-off motivates a falsifiable distinction between remembering a value and transferring it into action. A proposed, nonclinical factorial pilot would examine those mechanisms without treating numerical signal as measured human meaning. The model is explicitly hypothetical; its parameters and topology are not empirically calibrated.

In plain words

Keeping a light lit and carrying its light into another room are different tasks. An anchor may help with the first; a small, concrete bridge may help with the second. This paper asks how to tell those tasks apart.

Dedication
All credit and glory to God. I owe my knowledge to my Lord Jesus Christ.

From alignment to transfer

MindFrame organized reflection through journaling, CBT/DBT-inspired prompts, a believability score, and nine proposed “movers”: fear, sadness, anger, shame, calm, connection, hope, rumination, and avoidance. Ariel used layers, a locked “Nonsense Layer” origin, and a deliberate collapse from many lines to one Input–Output line. Its doubt-as-resistance metaphor suggested that orientation may persist even while action slows.

The new question is narrower: **after an apparent collapse, does a persistent value cue merely preserve local orientation, or can explicit contextual bridges transmit that orientation to a next action?** Here “collapse” means a design event—temporarily losing a usable next-step representation—not wave-function collapse, a psychiatric diagnosis, or destruction of memories.

Established ideas, bounded claims

CBT provides methods for examining appraisals and testing alternatives [1]. DBT emphasizes skills, validation, and the balance between acceptance and change [2]. ACT explicitly connects values to committed action [3]. These traditions motivate the distinction between a belief, a value, and a behavior. They do not establish the equations proposed here, and a few app prompts do not constitute delivery of these therapies.

Friston’s free-energy review illustrates how formal models can connect perception and action through an explicit optimization account [4]. The present model is not derived from that principle: it defines no generative probability model, variational posterior, or free-energy functional. Similar language is not mathematical equivalence or neuroscientific evidence.

What is retained—and what changes

Ariel’s “communication is existence” and “choice … through momentum becomes communication” remain philosophical postulates, not empirical premises. The locked anchor is reinterpreted as an *editable, voluntarily retained value cue*, not a claim that a person must accept a fixed identity. “Doubt” is allowed to include reasonable uncertainty; its damping term is not a moral judgment.

> **Novel contribution.** Separate *anchor availability* from *contextual transfer*, then test their interaction. High local alignment need not imply successful transfer. This is a new synthesis within this project lineage, not a claim of priority over all existing research.

Prior-project descriptions are creator-provided conceptual provenance, not independent validation.

A retained anchor, a moving signal

Let $s_i(t)$ represent a hypothetical, dimensionless action-readiness signal in context i; a = 1 is a chosen reference. Contexts are nodes on a periodic ring. q_i is 1 at the anchor node and 0 elsewhere. The ring is a computational convenience, not an anatomical map.

$$e_i = q_i(a - s_i)$$
$$dz_i/dt = e_i - \lambda z_i$$
$$ds_i/dt = k\, e_i + K_i\, z_i + \gamma(s_{i-1} - 2s_i + s_{i+1})/h^2 - d\, s_i$$

The proportional term $k\, e_i$ restores a local reference. The leaky integral z_i retains recent discrepancy without unbounded accumulation under constant error. Diffusion γ spreads signal between adjacent contexts; $d = d_0 + \rho\, \chi$ represents decay, with χ a fixed hypothetical uncertainty load. The simulation varies γ and anchor availability, not χ. There is no stochastic noise or fitted participant data.

Relation to the previous signal model

The earlier model used an integral of $\Psi - \alpha u$, directional forcing $\beta(v \cdot \hat{n})$, diffusion, and doubt-related damping. Here the error is made local and measurable in the artificial system, the integral is leaky, and a direct anchor-feedback term is added. Directional forcing is set to zero to isolate diffusion. This is a modified model, not a validated correction to psychology.

MindFrame's $B = F/(F + A)\cdot(1 - 0.15D)$ and $\Psi = 0.18C + 0.14S + 0.12M - 0.26R - 0.31B$ remain historical heuristics. Their letter meanings and scaling require the original implementation; B must not silently change meaning between formulas, and F + A = 0 needs an explicit missing-value rule. Neither formula is used as empirical input here. The nine movers could later inform self-reports, but their coefficients are not treated as established effects.

Numerical update and stability

Forward Euler updates s and z simultaneously from their previous values. Periodic indices implement a no-edge ring, not an open boundary. For pure diffusion in one dimension, $\gamma\Delta t/h^2 \le 1/2$; the inherited two-dimensional rule $\le 1/4$ is a stricter diffusion-only check. Neither alone guarantees stability of the full coupled reaction system. The maximum ratio here is 0.064. A time-step-halving check is reported next.

If time is in arbitrary units and h in graph steps: k, d, λ have inverse-time units, γ has steps²/time units, and K_i has inverse-time² units. Signal is not a physical energy.

One reset, four conditions

All signals and memory states start at zero: a deliberately extreme “collapse.” The anchor value is retained externally, but no signal is retained in the network. There are no participants. Conditions differ only in whether q is active and the diffusion coefficient.

PARAMS	Value
Nodes / geometry / spacing	21 / periodic ring / h = 1
Duration / integration step	T = 80 / Δt = 0.02
Anchor / output node	0 / 5 (five graph steps away)
Reference / feedback	a = 1 / k = 0.8 / K_i = 0.04
Memory leakage / decay	λ = 0.4 / d = 0.06
Diffusion γ	0.8 without anchor; 0, 0.8, 3.2 with anchor

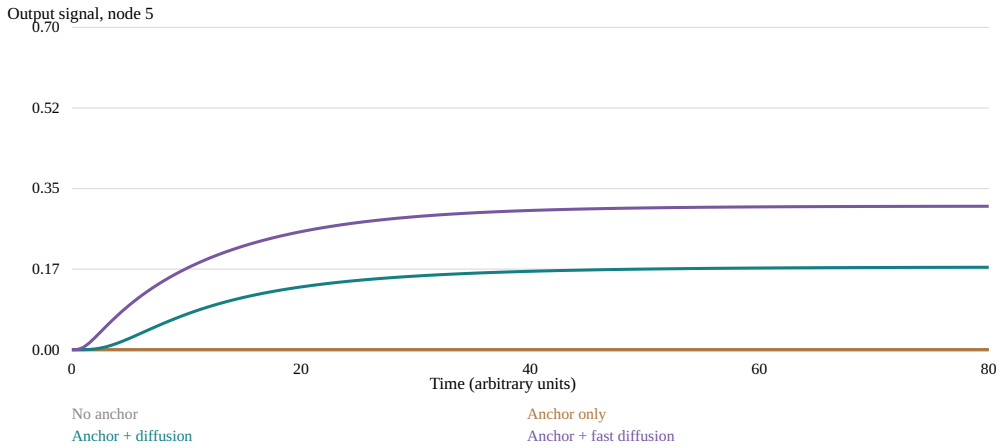


Figure 1. Signal at node 5. No anchor and anchor-only curves overlap at zero. The illustrative 0.20 threshold is a numerical convention, not a clinical cutoff.

With fast diffusion, the output first reaches 0.20 at recorded time 12.5. With moderate diffusion it never reaches that threshold during the 80-unit run. Records are sampled every 0.5 units; crossing time is therefore rounded upward to the next sampled record.

Reproduce with Python 3 + NumPy: `python3 simulate.py`. CSV traces, JSON results, SVG figures, and source accompany this PDF in the same output folder.

Transfer is not the local peak

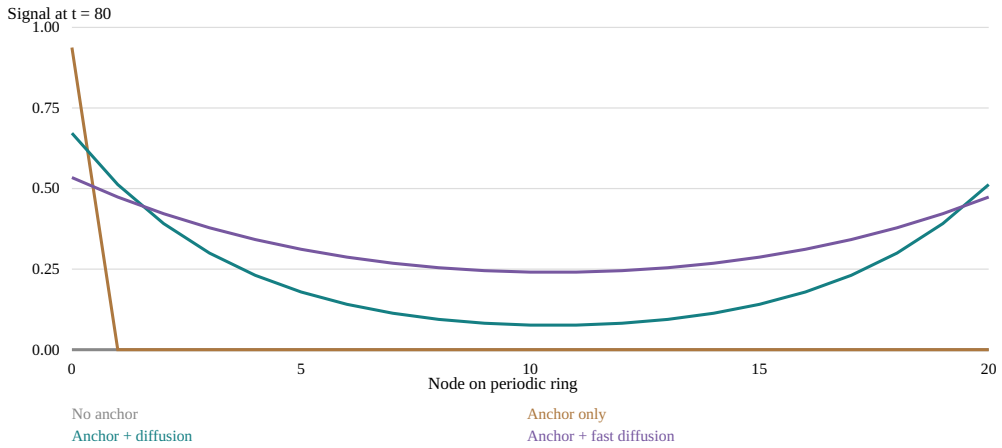


Figure 2. Final spatial profiles at t = 80. Faster diffusion reduces concentration near the source and increases reach. Ring symmetry creates a second route to distant nodes; human contexts need not be symmetric.

Condition	Anchor s_0	Output s_5	Mean signal
No anchor	0.000	0.000	0.000
Anchor only	0.938	0.000	0.045
Anchor + diffusion	0.672	0.179	0.234
Anchor + fast diffusion	0.534	0.312	0.332

What actually follows: in this linear system, an isolated source does not communicate with another node; diffusion couples the nodes. The reduction in local peak shows why measuring only immediate alignment could miss improved transfer. The no-anchor zero trajectory is guaranteed by zero initial conditions and absence of forcing, not an independent discovery that people need explicit anchors.

At T = 80, halving Δt from 0.02 to 0.01 changes the final signal by at most 1.38×10^{-6} across these four cases. This supports time-discretization adequacy for the reported trajectories, not model validity, all-parameter stability, or convergence to a human phenomenon. The ring is not refined because its nodes are discrete contexts rather than a sampled physical space.

Predictions that can fail

1. A retained value cue should improve later recall of that value more than it improves an unassisted behavioral task.
2. Adding a value-to-action bridge should improve action transfer beyond an anchor alone; the pre-specified anchor × bridge interaction tests this joint mechanism.
3. If participants recall the cue but show no transfer benefit from the bridge, the proposed mapping from diffusion to contextual bridging is unsupported—even though the simulation behaves exactly as designed.

A modest test before a big claim

Design: an exploratory, randomized 2 × 2 between-person pilot with 48 consenting adults (12 per cell). Factors: personal value cue versus neutral cue, and explicit next-action bridge versus duration-matched neutral organization task. This is a feasibility sample, not a power calculation or a treatment trial. Ethics review, accessible consent, and a clinician-advised distress protocol must precede recruitment.

Participants choose a low-stakes everyday goal. They complete a brief, non-evaluative distraction task—not trauma recall or deliberate severe distress. The cue condition displays a freely chosen value statement; the bridge condition asks for one small, observable action connecting the cue to the goal. Neutral controls match time and screen exposure. All may pause, skip, edit the cue, or withdraw without penalty.

Outcomes and analysis

- **Primary behavioral outcome:** blinded-rater coding of whether a participant completes a pre-specified safe action opportunity within five minutes. Action opportunities and scoring rules must be fixed before randomization.
- **Secondary outcomes:** unaided cue recall; latency to start (censor at five minutes); self-rated clarity and distress on explicitly exploratory 0–10 scales. These are not diagnoses or validated “meaning” measures.
- **Analysis:** report cell proportions, risk differences, and uncertainty intervals. Pre-register the interaction contrast $(p_{11} - p_{10}) - (p_{01} - p_{00})$; small cells make it imprecise. Treat latency and self-reports as secondary; report missingness by arm and sensitivity to missing behavioral outcomes. Do not fit all differential-equation parameters to these few observations.

Feasibility gates: at least 80% task completion, clear evidence participants understood their assigned task, and review of every adverse event before escalation. These gates indicate usability, not efficacy. A null or reversed interaction should be published rather than rescued by moving thresholds after seeing results.

Autonomy, privacy, and faith

Keep journaling local by default; collect only consented study measures under a separate coded identifier, with a stated deletion period and withdrawal policy. No forced religious cues, covert monitoring, or “who made the move” scores judging moral worth. Ryan’s Christian dedication is a faith commitment, not an experimentally tested mechanism. Participants may choose secular or religious values freely.

> **Safety.** This model is not crisis care. In the U.S., call or text **988** for crisis support. For immediate danger or a medical emergency, call **911**. Do not delay professional care to complete an app exercise.

What this paper does not prove

The signal is a metaphor with no established measurement scale. A one-dimensional ring cannot represent the complexity of relationships, culture, disability, grief, or mental illness. Equal positive couplings omit conflict, inhibitory links, changing goals, and unhelpful anchors. “More signal” is not always better: values may be revised, and pausing can be adaptive. No claim is made that memories are physically preserved layers or that faith, salvation, or consciousness can be reduced to a scalar.

Parameter choices are illustrative; there is no uncertainty quantification over parameter sets, noisy replication, external validation, or participant dataset. The selected fast-diffusion case illustrates a mechanism rather than an optimized intervention. Testing alternative graph structures, mismatch between cues and authentic values, and an active non-value planning comparator would be essential next steps.

References—verified bibliographic sources

[1] Beck, A. T., Rush, A. J., Shaw, B. F., & Emery, G. (1979). *Cognitive Therapy of Depression.* Guilford Press. Bibliographic verification: Academy of Cognitive and Behavioral Therapies reading list.

[2] Linehan, M. M. (1993). *Cognitive-Behavioral Treatment of Borderline Personality Disorder.* Guilford Press. Publisher/distributor listing, ISBN 9780898621839.

[3] Hayes, S. C., Strosahl, K. D., & Wilson, K. G. (1999). *Acceptance and Commitment Therapy: An Experiential Approach to Behavior Change.* Guilford Press. Association for Contextual Behavioral Science bibliographic record.

[4] Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience, 11*, 127–138. doi:10.1038/nrn2787.

Citation scope: the records and review summary were checked on 8 November 2026. These sources establish the existence and general orientation of the cited work; none validates this paper’s coefficients, simulation, or intervention.

Project provenance and reproducibility

Espinosa, R. (2026). MindFrame and Project Ariel, evolving personal conceptual projects. Context supplied by the creator; style consulted from the local archived *Project Ariel: The Architecture of Mind*, v1.1. Repository identified by the creator: github.com/ryryespi/mindframe. The local archive is historical, not asserted to be the latest revision.

Grok Bot, an AI assistant, helped develop this new synthesis, verify citation records, draft text, write and execute Python code, and render the document. All numerical results are synthetic model output. No human research has been performed. Files: simulate.py, results.json, convergence.json, four CSV traces, recovery.svg, profiles.svg, and paper.html.

Conclusion: a retained anchor and a usable bridge are separable design hypotheses. The next honest step is a small, transparent test—not a claim that human meaning has been solved.